• Second-order derivative test for local extrema

Theorem (Second-order Derivative Test)

Suppose the second-order partial derivatives of f are continuous in a disk centered at (a, b), and $\nabla f(a, b) = \mathbf{0}$, i.e., $f_x(a, b) = f_y(a, b) = 0$. Let

$$D = D(a, b) = f_{xx}(a, b) f_{yy}(a, b) - [f_{xy}(a, b)]^2 .$$

Then we have

(a) *If $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum.*

(b) *If $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum.*

(c) *If $D < 0$, then $f(a, b)$ is not a local extremum (a saddle point.)*

Remarks.

1. To remember the formula for the discriminant D, it is convenient to write it as a determinant $D = \det \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$.
2. In case (c), $D < 0$, (a, b) is a saddle point of f.
3. If $D = 0$, the test gives no information: f could be a local maximum, a local minimum at (a, b), or (a, b) could be a saddle point of f.

Example (Back to one of examples)

Classify all stationary points for $f(x, y) = y^2 - y^4 - x^2$.

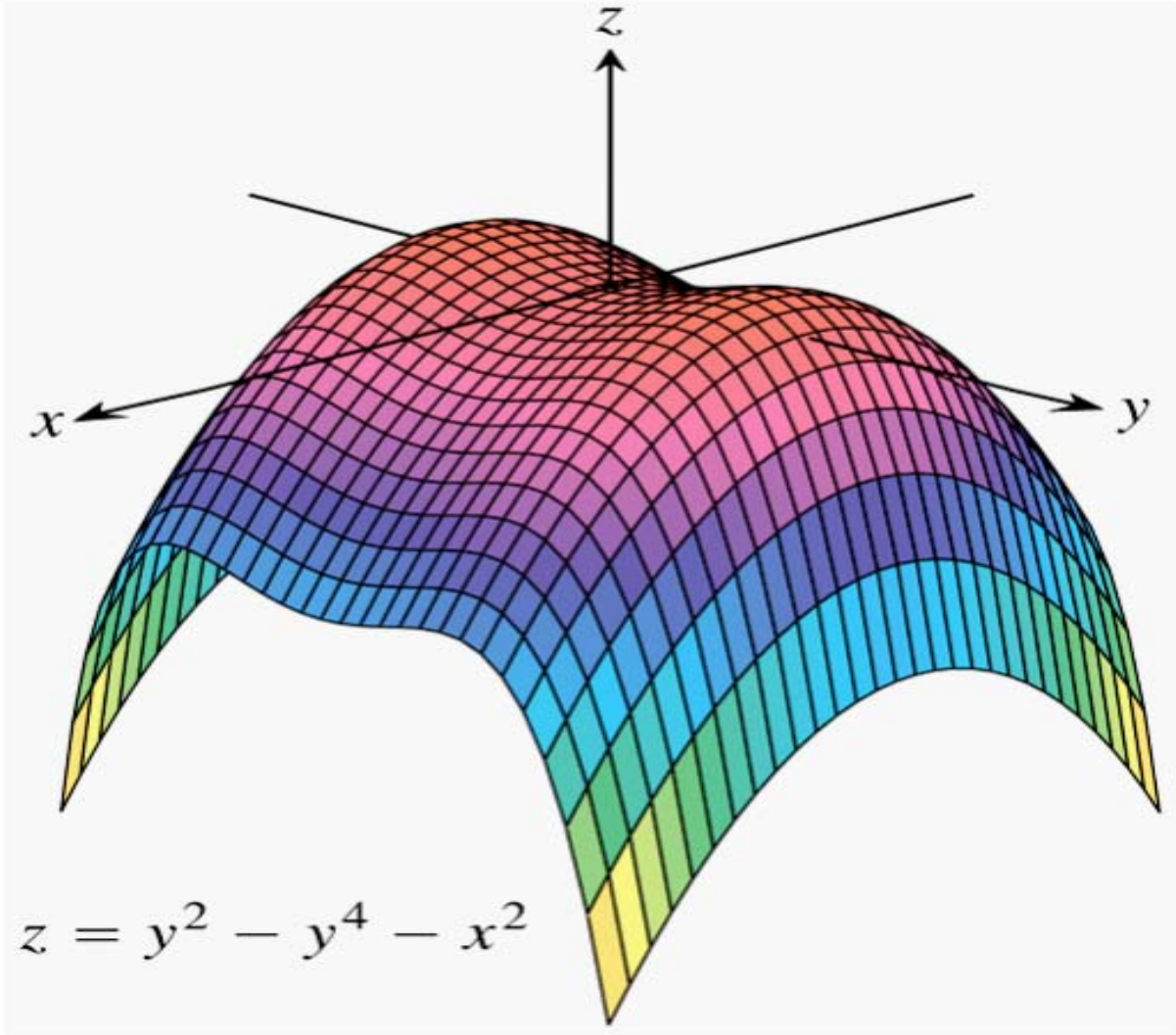
Solution. We have found above that stationary points of f are $(0, 0)$, $(0, \frac{1}{\sqrt{2}})$, and $(0, -\frac{1}{\sqrt{2}})$.

We use the Second-order Derivative Test to classify them. Since

$$f_{xx} = -2,\ f_{xy} = 0,\ f_{yx} = 0,\ f_{yy} = 2-12y^2 \implies D(x, y) = -2(2-12y^2).$$

- At $(0, 0)$ we have $D(0, 0) = -4 < 0$, and hence $(0, 0)$ is a saddle point.
- At $(0, \pm\frac{1}{\sqrt{2}})$ we have $D(0, \pm\frac{1}{\sqrt{2}}) = -2(2 - 6) = 4 > 0$, and $f_{xx} = -2 < 0$. Then $f(0, \pm\dfrac{1}{\sqrt{2}})$ are local maximum.

- Diagram: $f = y^2 - y^4 - x^2$



Example

Find the local extrema of $f(x, y) = x^4 + y^4 - 4xy + 1$.

Solution. Note that $\nabla f(x, y) = (4x^3 - 4y, 4y^3 - 4x)$. Then

$$\nabla f(x, y) = \mathbf{0} \iff \begin{cases} x^3 - y = 0 \\ y^3 - x = 0 \end{cases} \iff \begin{cases} y = x^3 \\ x = y^3. \end{cases}$$

So we have $x = x^9$ and hence

$$\begin{aligned} 0 &= x^9 - x = x(x^8 - 1) = x(x^4 - 1)(x^4 + 1) \\ &= x(x - 1)(x + 1)(x^2 + 1)(x^4 + 1), \end{aligned}$$

which gives $x = 0$, $x = 1$ or $x = -1$.

Thus there are three critical points: $(0, 0)$,$(1, 1)$, and $(-1, -1)$.

Example (cont.)

The discriminant of second-order derivatives is

$$D(x,y) = f_{xx}f_{yy} - [f_{xy}]^2 = (12x^2)\cdot(12y^2) - (-4)^2 = 144x^2y^2 - 16.$$

- Since $D(0,0) = -16 < 0$, it follows that $(0,0)$ is a saddle point.
- Since $D(1,1) = 128 > 0$ and $f_{xx}(1,1) > 0$, it follows that $f(1,1) = -1$ is a local minimum.
- Since $D(-1,-1) = 128 > 0$ and $f_{xx}(-1,-1) > 0$, it follows that $f(-1,-1) = -1$ is also a local minimum.

Example

Find the local extrema of $f(x,y) = 3y^2 - 2y^3 - 3x^2 + 6xy$.

Solution. We have $\nabla f(x,y) = (6y - 6x, 6y - 6y^2 + 6x)$. Then

$$\nabla f(x,y) = \mathbf{0} \iff \begin{cases} 6y - 6x = 0 \\ 6y - 6y^2 + 6x = 0 \end{cases} \iff \begin{cases} y = x \\ 6x(2-x) = 0. \end{cases}$$

So we obtain two points $(0,0)$ and $(2,2)$, which are critical points.

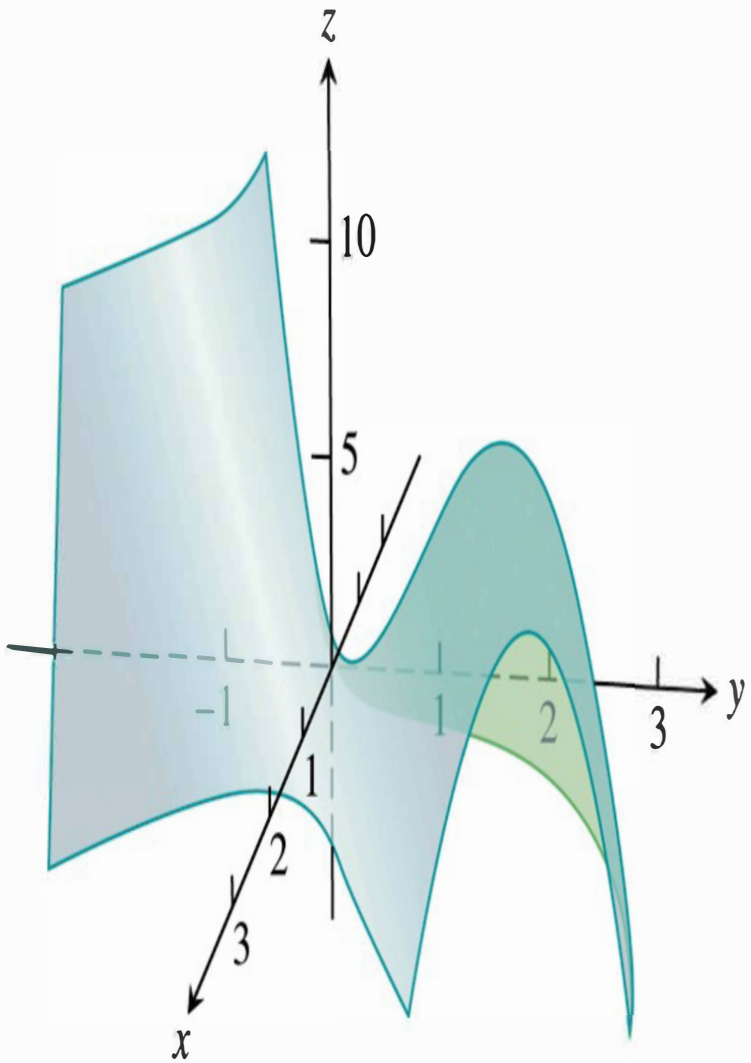
Now we calculate the second derivatives

$$\begin{cases} f_{xx} = -6 \\ f_{yy} = 6 - 12y \\ f_{xy} = f_{yx} = 6 \end{cases} \quad \Longrightarrow \quad D = f_{xx} f_{yy} - (f_{xy})^2 = 72(y-1).$$

At the critical point $(0,0)$, we have $D = -72 < 0$ which implies that the function has a saddle point at the origin.

At the critical point $(2,2)$, we have $D = 72 > 0$. Since $f_{xx} = -6 < 0$, we conclude that the function has a local maximum at $(2,2)$ and its value is $f(2,2) = 8$.

- Diagram: $f = 3y^2 - 2y^3 - 3x^2 + 6xy$



• Global Maximum and Minimum Values

Definition

(a) A function f of two variables has a global (absolute) maximum on its domain D at a point (a, b) if

$$f(x, y) \leq f(a, b), \quad \text{for all points } (x, y) \in D.$$

The number $f(a, b)$ is called a global (absolute) maximum value.

(b) If the inequality above is replaced by

$$f(x, y) \geq f(a, b), \quad \text{for all points } (x, y) \in D,$$

then f has a global (absolute) minimum on D at a point (a, b), and $f(a, b)$ is called a global (absolute) minimum value.

Example

Find the global minimum of $f(x, y) = x^2 + y^2 - 2x - 6y + 14$ on $\mathbb{R}^2$.

Solution. Completing the squares for f, we have

$$f(x, y) = 4 + (x - 1)^2 + (y - 3)^2 \geq 4, \quad \text{for all } x \text{ and } y.$$

Equality occurs when $(x - 1)^2 = (y - 3)^2 = 0 \iff x = 1, y = 3$.

Then the value $f(1, 3) = 4$ is the global minimum of f.

Note that $f(x, y)$ does not have a global maximum.

• Existence of global Max/Min values

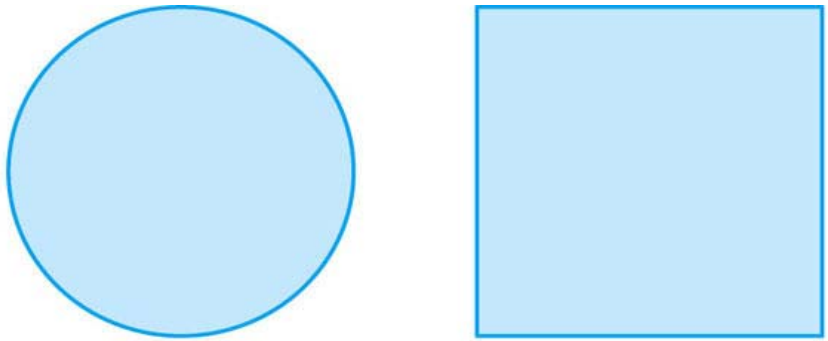
In general, the global maximum or minimum of a function may not exist.

However, when f is continuous and the domain satisfies some nice conditions, the existence of global extrema is guaranteed.

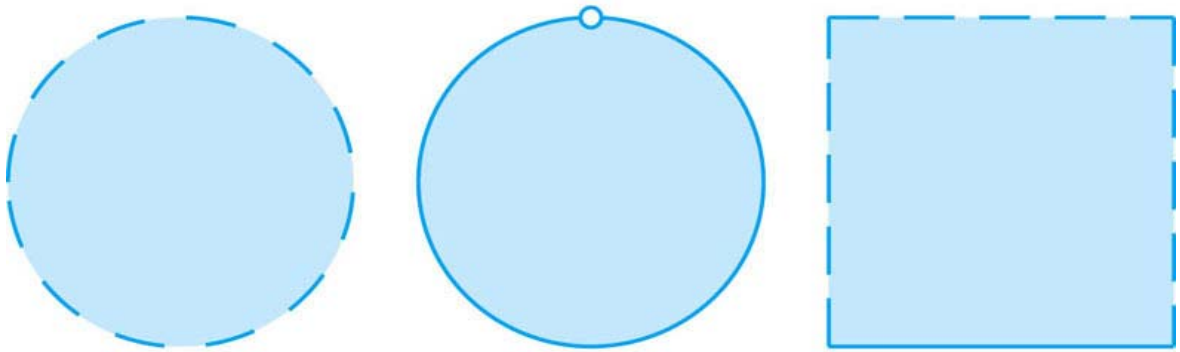
Theorem

Suppose $f(x, y)$ is continuous in its domain D, where D is closed and bounded. Then f has global maximum and global minimum.

• Examples of closed sets



(a) Closed sets



(b) Sets that are not closed

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Outline

1. Directional Derivatives
2. Maximum & Minimum Rate of Change
3. Optimization Problems
4. Double Integrals

• Optimization Problems

In optimization problems we are looking for the largest value or the smallest value that a function can take.

For example,

- Find the shortest distance from the point $(1, 2)$ to the curve $xy = 4$.
- Suppose the temperature on a hot plate is $T(x, y) = x - y$. Find the hottest spot on the ellipse $2x^2 + 9y^2 = 16$.
- What is the least surface area of a closed right circular cylindrical can whose volume is 18π cm^3?
- A rectangular box without a lid is to be made from $12m^2$ cardboard. Find the maximum volume of such a box.

• Lagrange Multiplier - One constraint

We discuss the method of Lagrange multipliers for maximizing or minimizing a function $f(x, y)$ subject to a constraint of the form $g(x, y) = k$. That is

$$\begin{cases} f(x, y) & \longrightarrow & \text{max/min} \\ g(x, y) = k. \end{cases}$$

• Lagrange Multipliers for $f(x, y)$

ASSUMPTION: Global extreme values exist.

Algorithm (Lagrange Multipliers for $f(x, y)$)

To find the maximum and minimum values of $f(x, y)$ subject to the constraint $g(x, y) = k$, we do the following steps:

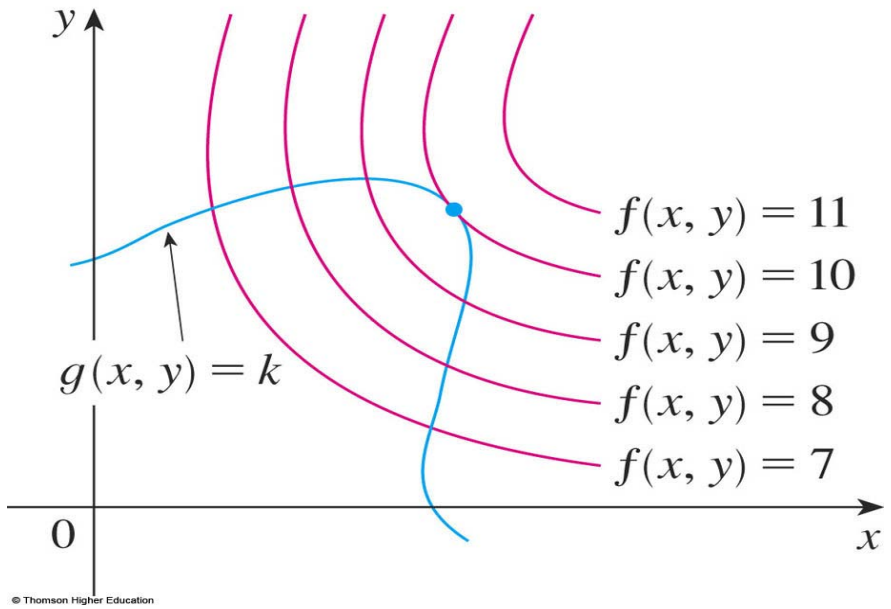
Step 1. Find all values of x, y, and λ satisfying the equations

$$\nabla f(x, y) = \lambda \nabla g(x, y) \quad \text{and} \quad g(x, y) = k.$$

Step 2. Evaluate f at all points (x, y) that are found from Step 1. The largest of these values is the maximum value, while the smallest of these values is the minimum value.

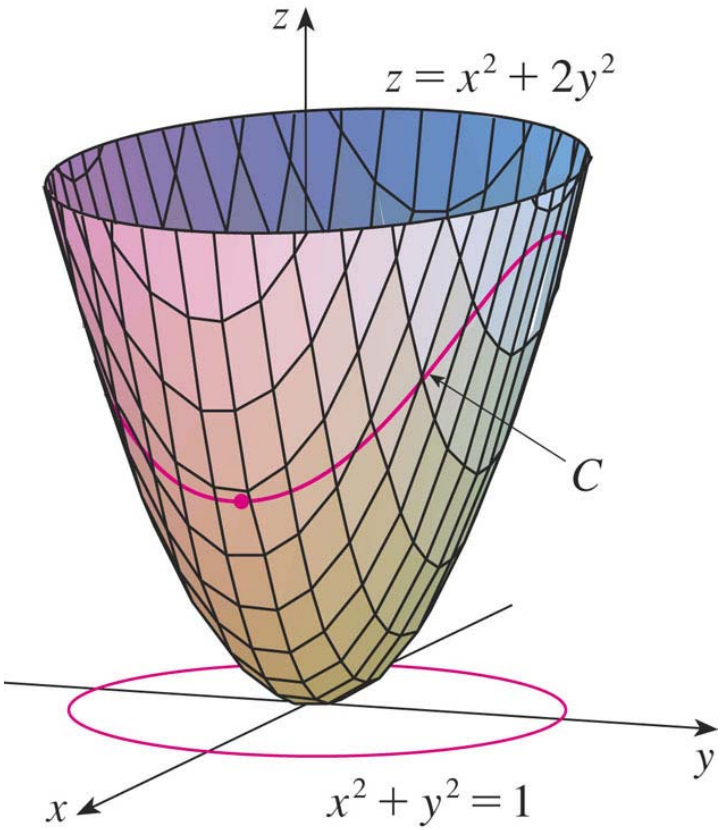
- **Remark.**
  1. If $\lambda = 0$, then $\nabla f(x, y)$ gives stationary points of f.
  2. If $\lambda \neq 0$, then both vectors $\nabla f(x, y)$ and $g(x, y)$ are parallel or opposite each to other.

- **Diagram:** $\nabla f(x, y) = \lambda \nabla g(x, y)$

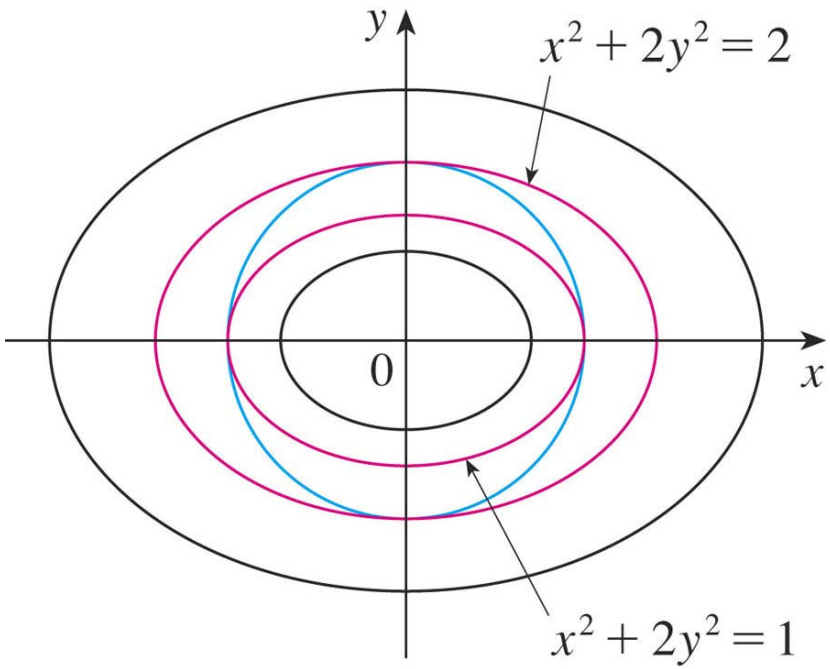


Example

Find the extreme values of the function $f(x, y) = x^2 + 2y^2$ on the circle $x^2 + y^2 = 1$.



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Example (cont.)

Solution. The circle $x^2 + y^2 = 1$ is closed and bounded, and the function $f(x, y) = x^2 + 2y^2$ is continuous everywhere on $\mathbb{R}^2$. Then the maximum and minimum values of f exist.

We have the problem $\begin{cases} f(x, y) = x^2 + 2y^2 & \longrightarrow \quad \max / \min \\ g(x, y) = x^2 + y^2 = 1. \end{cases}$

We apply the method of Lagrange multipliers which means to solve the equations

$$\nabla f = \lambda \nabla g \quad \text{and} \quad g(x, y) = 1.$$

Since $\nabla f = \lambda \nabla g$ means $f_x = \lambda g_x$ and $f_y = \lambda g_y$, we have

$$2x = 2x\lambda \tag{1}$$

$$4y = 2y\lambda \tag{2}$$

$$x^2 + y^2 = 1. \tag{3}$$

Example (cont.)

From (1) $2x = 2x\lambda$, we have $x(1-\lambda) = 0$, which gives $x = 0$ or $\lambda = 1$.
- If $x = 0$, then (3) $x^2 + y^2 = 1$ gives $y = \pm 1$.
- If $\lambda = 1$, then (2) $4y = 2y\lambda$ and we have $2y = 0 \iff y = 0$, and hence by (3) $x = \pm 1$.

Thus the maximum and minimum of f are at the following fours points $(0, \pm 1)$, $(\pm 1, 0)$.

Evaluating f at these points, we have $f(0, \pm 1) = 2$, $f(\pm 1, 0) = 1$.

Consequently, the maximum value of f on the circle $x^2 + y^2 = 1$ is 2, attained at $(0, \pm 1)$, while the minimum value of f on this circle is 1, attained at $(\pm 1, 0)$.

• Lagrange Multipliers for $f(x, y, z)$

The Algorithm is the same as for functions of two variables, adding the third variable z.

ASSUMPTION: Global extreme values exist.

Algorithm (Lagrange Multipliers for $f(x, y, z)$)

To find the maximum and minimum values of $f(x, y, z)$ subject to the constraint $g(x, y, z) = k$, we do the following steps:

Step 1. Find all values of x, y, z, and λ satisfying the conditions

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z) \quad \text{and} \quad g(x, y, z) = k.$$

Step 2. Evaluate f at all points (x, y, z) that are found from Step 1. The largest of these values is the maximum value, while the smallest of these values is the minimum value.

Example

A rectangular box without a lid is to be made from $12m^2$ cardboard. Find the maximum volume of such a box.

Solution. Let x, y, z denote respectively the length, width and height of the box ($x > 0, y > 0, z > 0$). Then the box's volume $V = xyz$ and the box's area (without a lid) is $g(x, y, z) = 2xz + 2yz + xy = 12$.

We have the problem $\begin{cases} V(x, y, z) = xyz \quad \longrightarrow \quad \max \\ g(x, y, z) = 2xz + 2yz + xy = 12. \end{cases}$

We solve equations $\nabla V = \lambda \nabla g$ and $g(x, y, z) = 12$, which become

$$yz = \lambda(2z + y) \tag{4}$$

$$xz = \lambda(2z + x) \tag{5}$$

$$xy = \lambda(2x + 2y) \tag{6}$$

$$2xz + 2yz + xy = 12. \tag{7}$$

Example (cont.)

Multiplying (4) by x, (5) by y, and (6) by z, we have

$$xyz = \lambda(2xz + xy) \tag{8}$$

$$xyz = \lambda(2yz + xy) \tag{9}$$

$$xyz = \lambda(2xz + 2yz). \tag{10}$$

It is important to note that $\lambda \neq 0$, for otherwise, $yz = xz = xy = 0$ from (4), (5), and (6), which would contradict (7).

From (8) and (9), we have $2xz + xy = 2yz + xy$, which gives $x = y$.

From (9) and (10), we have $2yz + xy = 2xz + 2yz$, which yields $y = 2z$.

Example (cont.)

Now substituting $x = y = 2z$ into (7) (which is $2xz + 2yz + xy = 12$), we have

$$4z^2 + 4z^2 + 4z^2 = 12 \iff z^2 = 1 \iff z = 1,$$

which gives $x = y = 2$.

This is the only point we found, and $V(2, 2, 1) = 4$.

To check whether 4 is max or min, we just take any point (x, y, z) with $g(x, y, z) = 12$ and compute $V(x, y, z)$ to compare with 4.

$\Longrightarrow$ Take say $\left(2, 1, \frac{5}{3}\right)$, for which $V\left(2, 1, \frac{5}{3}\right) = \frac{10}{3} < 4$. So we conclude that $V(2, 2, 1) = 4$ is the maximum volume.

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• Double Integrals

- Recall: Definite integral $\int_a^b f(x)\,dx$

  Let f be a continuous function on $[a, b]$.

  We partition the interval $[a, b]$ into n subintervals of equal length $\Delta x = \dfrac{b-a}{n}$ by $x_0 = a < x_1 < x_2 < \ldots < x_n = b$.

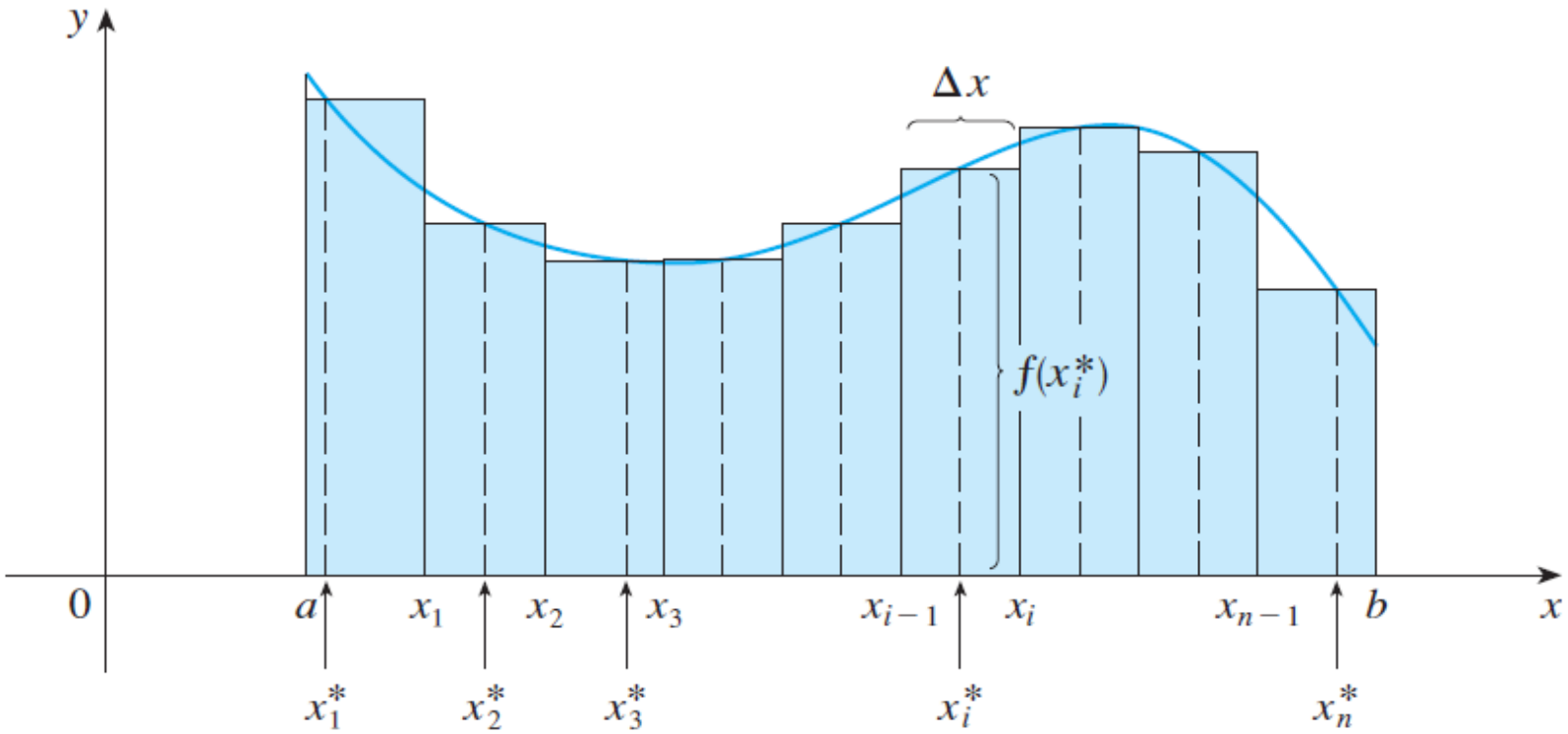
  We also let $x_1^*, x_2^*, \ldots, x_n^*$ be sample points from each subinterval, so $x_k^* \in [x_{k-1}, x_k]$.

Then the definite integral of f from a to b is

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x.$$

(independent of a choice of sample points (x_k^*)).

The expression $\sum_{k=1}^{n} f(x_k^*)\Delta x$ is called a **Riemann sum**.

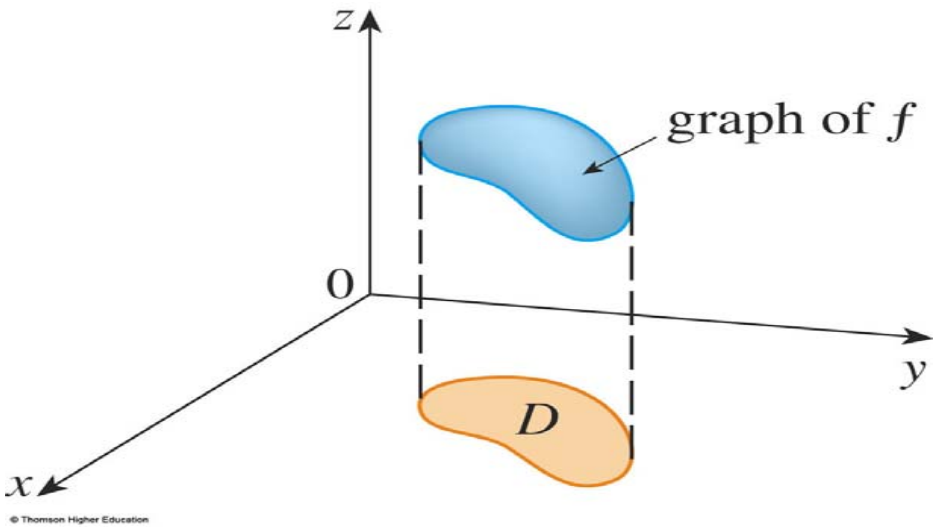

y
Δx
f(x_i^*)
0
a
x_1
x_2
x_3
x_{i-1}
x_i
x_{n-1}
b
x
x_1^*
x_2^*
x_3^*
x_i^*
x_n^*

• Double Integrals

The idea of a definite integral is generalized to double integrals of functions of two variables.

• **Our aim:** To study the integral of a function of two variables $f(P)$ over a region D in the plane $\mathbb{R}^2$, with the notation $\iint_D f(P)\, dA$.

• **Diagram:** Double integral



• Application: Volume under a surface over a region D

Consider a solid over a region D of the xy-plane.

Suppose $f(P) \geq 0$ is the height at each point $P \in D$, where f is a continuous function on D.

Application 1.

The volume V of the solid is given by

$$V = \iint_D f(P)\, dA.$$

Note that the solid lies above D and under the surface $z = f(P)$.

Remark. In particular, when $f(P) = 1$ on D, we obtain the area $A(D)$ of the region D as

$$A(D) = \iint_D 1\, dA.$$

• Application: Total charge, total mass, and moment of a lamina

Suppose a lamina (thin plate) occupies a region D of $\mathbb{R}^2$.

• An electric charge is distributed over D and the charge density (in units of charge per unit area) is given by $\sigma(P)$ at a point P in D, where σ is a continuous function on D.

Application 2.

The total charge Q is given by

$$Q = \iint_D \sigma(P)\, dA.$$

• The density (in units of mass per unit area) is given by $\rho(x, y)$ at a point (x, y) in D, where ρ is a continuous function on D.

Application 3.

(a) The total mass M is given by

$$M = \iint_D \rho(x, y)\, dA.$$

(b) The moment of the entire lamina about the x-axis, respectively y-axis, is

$$M_x = \iint_D y\rho(x, y)\, dA, \quad \text{respectively} \quad M_y = \iint_D x\rho(x, y)\, dA.$$

(c) The coordinates $(\bar{x}, \bar{y})$ of the center of the mass of the lamina are

$$\bar{x} = \frac{M_y}{M}, \quad \bar{y} = \frac{M_x}{M}.$$